Lecture 6: Cancer Cell Lines Encyclopedia (CCLE)

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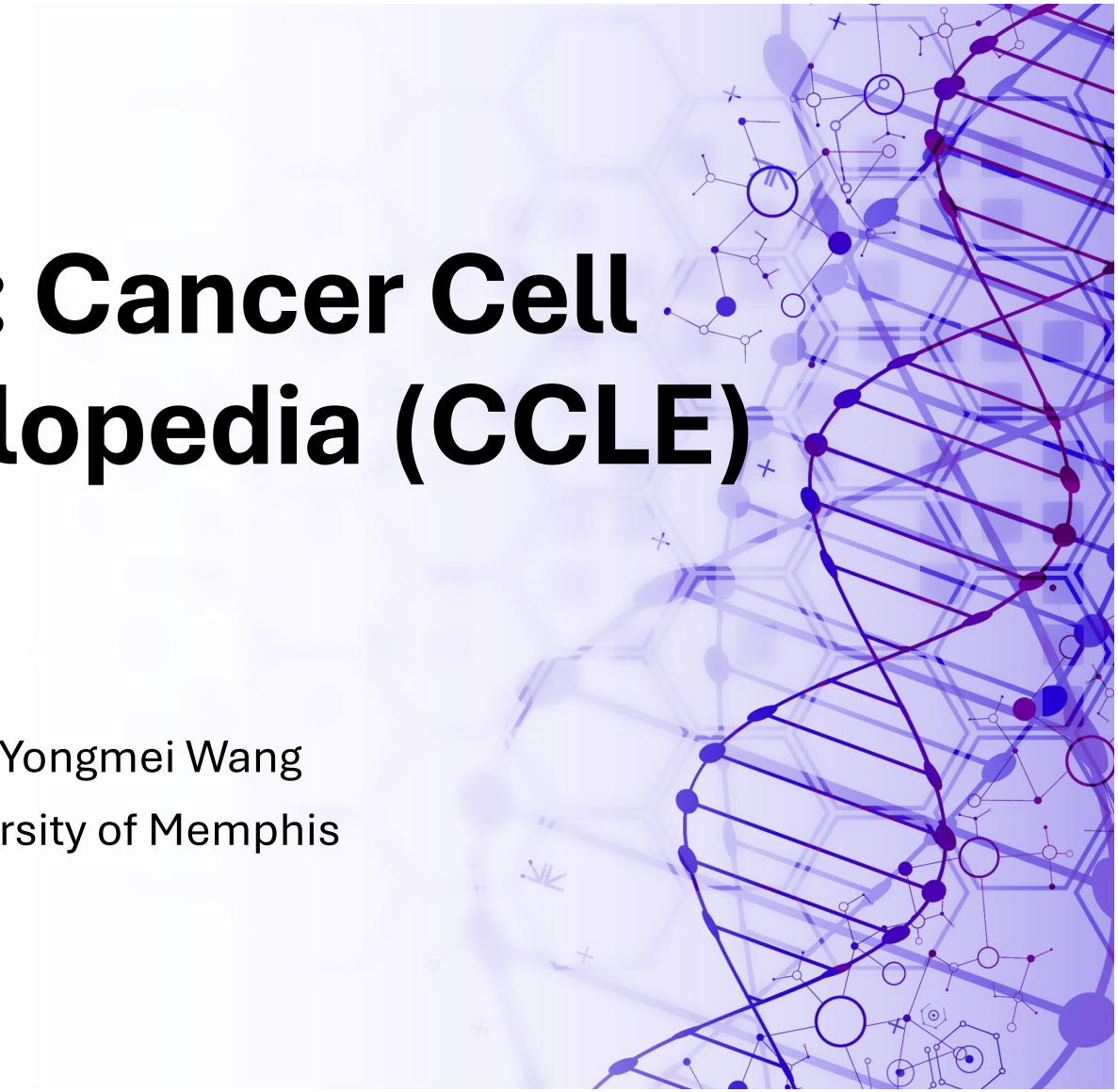


What is a cancer cell line?

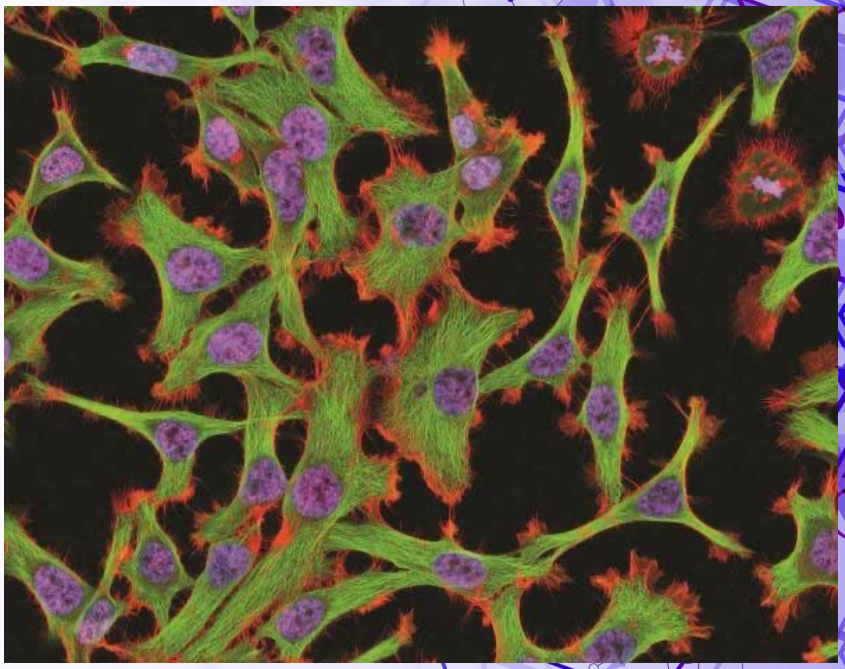
- A **cancer cell line** is a population of cancer cells that:
- **Originated from a patient's tumor**, and
- Has been grown **in the lab** under controlled conditions, so it can divide indefinitely.

Why Can It Be Grown Forever?

- Normal cells stop dividing after a few rounds (called **senescence**).
- Cancer cells often have mutations that allow them to **bypass normal limits on growth**.
- These cells can grow in petri dishes or flasks, a process known as **cell culture**.

History of first cancer cell line: HeLa cell

- First successfully grew in 1951
- Source: derived from a cervical cancer tumor of Henrietta Lacks, a young African American woman passed away from the disease that year
- Location: Johns Hopkins Hospital, Baltimore, MD.
- Scientist: Dr. George Gey.
- First human cells grown to divide indefinitely.
- Revolutionized biomedical research.



Why use cell lines?

- Easy to grow and manipulate in the lab.
- Represent various human cancers (e.g., breast, lung, colon).
- Serve as models to:
  - Study tumor genetics
  - Discover cancer vulnerabilities
  - Test therapeutic compounds

Cancer cell lines

- Used to study **Molecular Mechanism of Cancer cells**
- Study **signaling pathways** in cancer (e.g., PI3K/AKT, RAS/MAPK, Wnt).
- Knockdown or overexpress genes to observe functional changes.
- Investigate the role of **oncogenes** (e.g., KRAS, MYC) or **tumor suppressors** (e.g., TP53, RB1).

Modeling Specific Cancer Subtypes

- Cancers are heterogeneous diseases. For example, breast cancers have different subtypes.
- Subtypes are defined based on the presence or absence of certain **receptors** and **gene expression profiles**
- **Main breast cancer subtypes based on the presence of receptors**
  - **Luminal A (ER+, PR+, HER2-): best prognosis**
  - **Luminal B (ER+, PR+-, HER+-), moderate**
  - **HER2+ (ER-, PR-, HER2+)**
  - **Triple negative (ER-, PR-, HER2-); poor prognosis**

Discovery of Breast Cancer Subtypes

- In 2000, Perout et al published in Nature (https://www.nature.com/articles/35021093)
  - Used microarrays to profile ~65 tumors.
  - Identified molecular subtypes: Luminal A, Luminal B, HER2-enriched, Basal-like, Normal-like.
  - Introduced the concept of **intrinsic subtypes** based on gene expression.
  - They used 2000 genes in the test
- **Impact**: Revolutionized understanding of breast cancer heterogeneity.

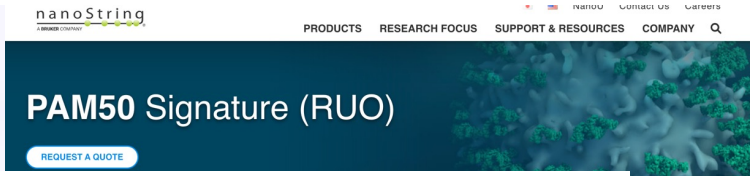
Development of PAM50 Classifier

- **2009 – Parker et al., in Journal of Clinical Oncology (https://pmc.ncbi.nlm.nih.gov/articles/PMC2667820/)**
  - Refined Perou's gene set to 50 genes (PAM50) for clinical use.
  - Classified tumors into the same 5 subtypes using qRT-PCR/NanoString.
  - These subtypes can guide the choice of therapy
  - Also introduced **Risk of Recurrence (ROR) score**.
- Led to the clinical test **Prosigna®**, used in ER+/HER2- breast cancer.
- Prosigna® diagnostic assay was developed and commercialized by NanoString Technologies, Inc .
- Dr. Charles Perou, now a professor at UNC-Chapel is the co-inventor for Prosigna®

Nanostrings Technologies Inc

- A biotech company, founded in 2003, under the leadership of Dr. Leroy Hood (spun off as an independent company)
- Technology platforms:
- **nCounter® : a digital g**ene expression platform that uses **color-coded molecular barcodes** to directly count RNA, DNA
- **Prosigna® Assay:** A **FDA-cleared diagnostic test** for early-stage, hormone receptor–positive breast cancer based on the **PAM50 gene signature**.
- **GeoMx® Digital Spatial Profiler (DSP):** A **spatial transcriptomics** and **proteomics** platform launched in 2019

Nanostrings Technologies Inc



- began in 2003 with early seed funding (~$3.8M).
- In 2012, a crucial **Series E of $15.3M** helped fund Prosigna® development.
- IPO in mid-2013 raised **$54M**, culminating in about **$82M** total funding before becoming a public company.
- 2021-2023 : faced lawsuits from 10xGenomics
- 2024: Filed for Chapter 11 bankruptcy, assets sold off
- **Bruker Corporation**, a global leader in life-science instrumentation, acquired its assets at $392.6 million (nCounter®, GeoMx®, CosMx™, AtoMx™, Prosigna®)
- Bruker aims to integrate NanoString's gene expression and spatial biology platforms into its broader **CellScape™** proteomics systems

What is CCLE?

- A public resource of cancer cell lines developed by the Broad Institute.
- Contains multi-omics data (genomic, transcriptomic, proteomic, **pharmacologic**).
- Used to study cancer biology and predict drug sensitivity.

https://sites.broadinstitute.org/ccle/

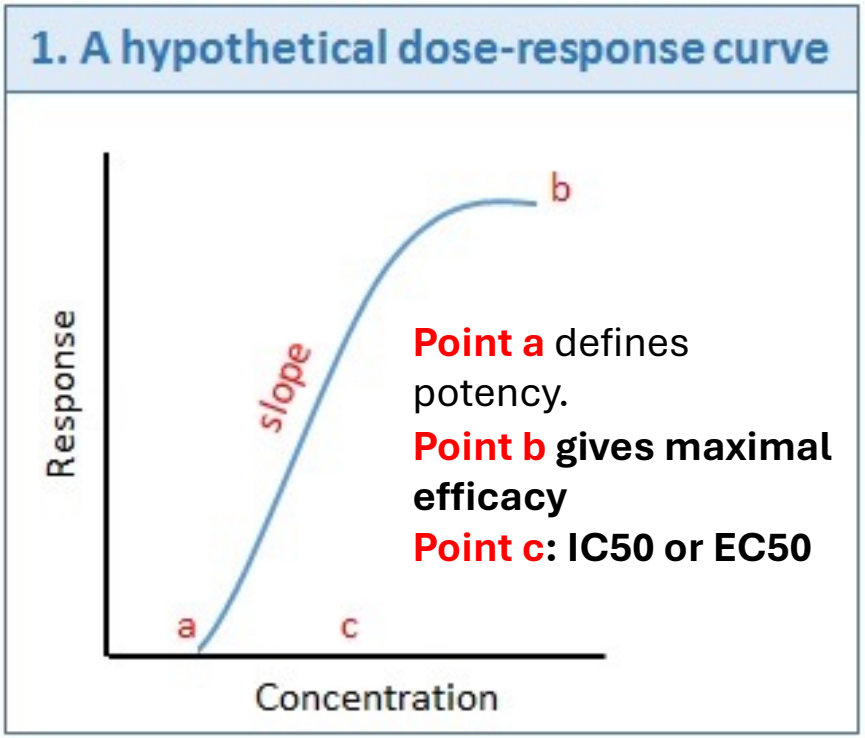
In-class Exercise

- Download “Model.csv” from DepMap Portal under “All Data”
- Open this “Model.csv” via jupyter notebook and find how many cell lines in the cancer you are interested in
- We will first look at breast cancer. You repeat again with other cancers

Drug screening using cancer cell lines

- TO evaluate the **efficacy, potency, and selectivity** of drugs against different cancer types.
- First, select cell lines
- Seed cell line into **multi-well plates** (96-well, 384-well). Allow cells adhere and grow to a certain density under standard conditions.
- A library of **candidate drugs** (or concentrations of a drug) is applied to cells.
- Cells are incubated with drugs for **24–72 hours** (or more)
- Assay Readouts (viability, apoptosis, and proliferation) and data analysis: Generate Dose-Response Curves

What is drug dose response curve?



1. A hypothetical dose-response curve

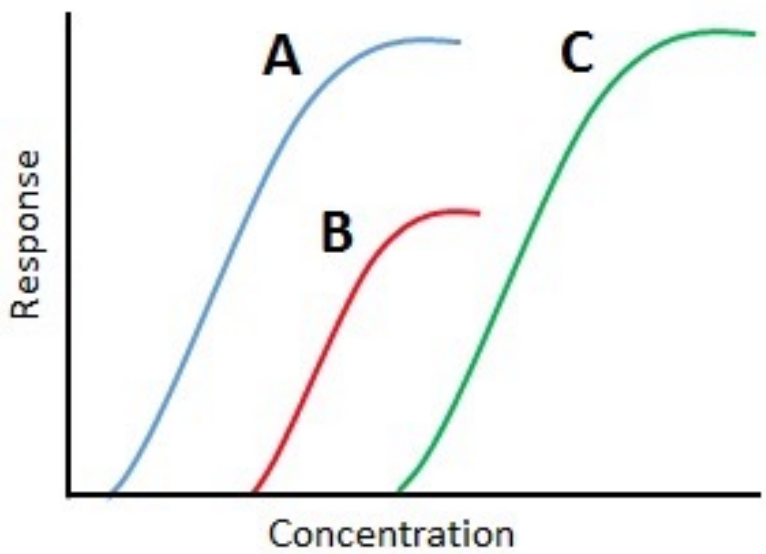
Dose-response graphs typically have the dose on the X-axis and the measured effect (i.e. the measured response) on the y- axis. Plotting the logarithm of the concentration generally results in sigmoidal plots as shown on the left.

The main parameters that can be identified by dose-response curves are

- Potency- the position of the curve along the x-axis **a**
- Maximal efficacy- the greatest response attainable **b**
- Slope- change in response per unit dose (and half maximal dose **c**)

https://www.pharmacologyeducation.org/pharmacology/dose-response-relationships

2. Comparison of dose-response curves



2. Comparison of dose-response curves

The pharmacologic profiles of individual drugs can be differentiated by comparing their dose-response curves.

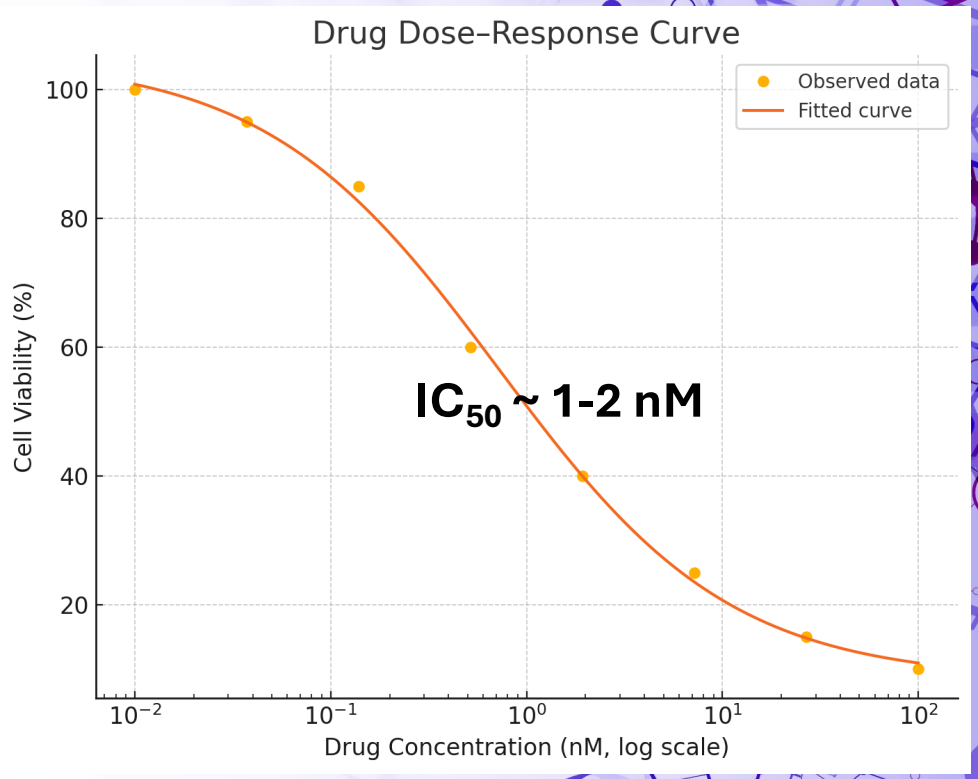
In the graph on the left, drug **A** has greater biological activity per dosing unit and is therefore considered to be more potent than drugs **B** or **C**- shown by its left-shifted position on the x-axis. Drugs **A** and **C** have equal efficacy- indicated by their maximal attainable response (ceiling effect). Drug **B** is more potent than drug **C**, but its maximal efficacy is lower.

Drug A is more potent than B and C
Drug A and C have greater maximum efficacy than B

https://www.pharmacologyeducation.org/pharmacology/dose-response-relationships

Example: a drug is tested on a cell line

Concentration (nM)	% Cell Viability
0.01	100% (no effect)
0.1	95%
1.0	60%
10.0	25%
100.0	10%



What drug sensitivity data available?

- **The Cancer Cell Line Encyclopedia (CCLE)**, through the **DepMap portal**, provides drug sensitivity data from several large-scale screening projects.
- **PRISM Repurposing Dataset (Broad Institute)**
  - Pooled barcoded screening of ~500 cell lines
  - >4,500 compounds (many FDA-approved or investigational)
  - Relative viability (log fold change) at multiple concentrations
  - Released in **2019–2021;**
  - **second screening was done with multiple doses; first screening was done with one dose**

- **Genomics of Drug Sensitivity in Cancer (**Sanger Institute)
  - ~1,000 human cancer cell lines from various tissues
  - ~500 anti-cancer drugs (including targeted therapies and chemotherapies)
  - GDSC1: Early screen using fewer drugs.
  - GDSC2: Newer dataset using updated assay and more robust analysis.
  - Links **drug sensitivity** (like IC50) to **genomic features** such as mutations, copy number variation, gene expression, and methylation.
  - Between **2010 and 2018**.

- **CTD^2 Dataset (Cancer Target Discovery and Development)**
  - This is more recent collaborative effort; Encourage to find out more about this on your own

Breast cancer drugs on the market

- Chemotherapy: doxorubicin; paclitaxel,
- Hormone therapy: tamoxifen, Anastrozole (Arimidex), exemestane (Aromasin)
- Targeted therapy: trastuzumab, pertuzumab, and palbociclib
- New developments include targeted therapy, immunotherapy, metastasis detections

Exercise

- Download drug screening data from DepMap Portal
- Compare different drugs potency in different breast cancer cell lines by plotting dose-response curves
- This exercise did not get finalized because the data were very noisy